

Non-Planar FDM Printing

Advanced Digital Fabrication | Task 4 | Mackey (Ho Yin) Tsang



Contents

STAGE 0

01

Introduction

02

Z Anti-Aliasing

Orcu Slicer

03

Bricklayer

PrusaSlicer (Official)

STAGE 1

04

Non-Planar Slicing

PrusaSlicer 2.6

05

Transparency Test

Prusa Non-Planar

06

Application

Prusa Non-Planar

07

Conclusion

01

Introduction

Non-planar slicing has been an active research and hobbyist topic for over a decade, with diverse algorithms developed by researchers and open-source contributors. FDM application remains limited: toolhead geometry differs per machine, requiring individual characterisation. This project applies these methods to Digital Research Hub (DRH) hardware, evaluating viability for research and education.



Fig 1.1 - Mitigation of staircase effect by implementation of nonplanar slicing in Slic3r [5]

DRH Hardware



Bambu Lab A1 mini
Z Anti-Aliasing (Chaper 02)



Purusa XL (5 Toolhead)
Bricklayers (Chapter 03)
Non-Planar Slicing (Chapter04-06)

02

Z Anti-Aliasing (ZAA)

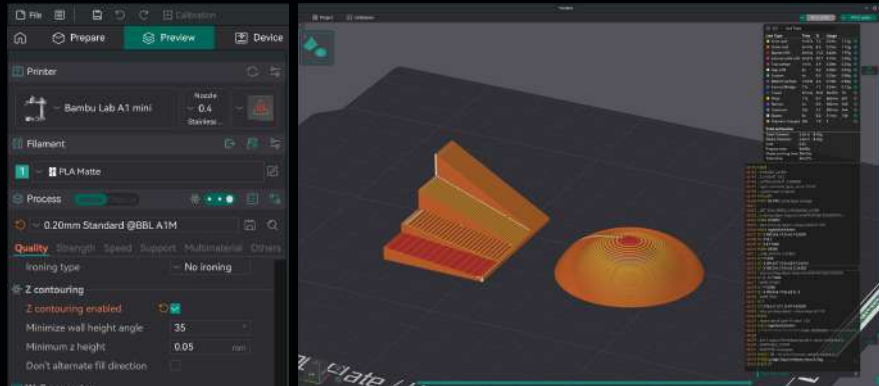


Fig 2.1 - OrcaSlicer with Z Contour Enabled

BambuStudio-ZAA [2], a micro-non-planar method since merged into OrcaSlicer, was tested on the Bambu Lab A1 mini. Stair (5/10/20/30°) and dome (10/20/40/60°) geometries were printed. Improvement was tactile at 5°, visible to 10–20°, negligible beyond 30°. Inter-layer gaps increased with ZAA active, limiting application to parts intended for post-process sanding.

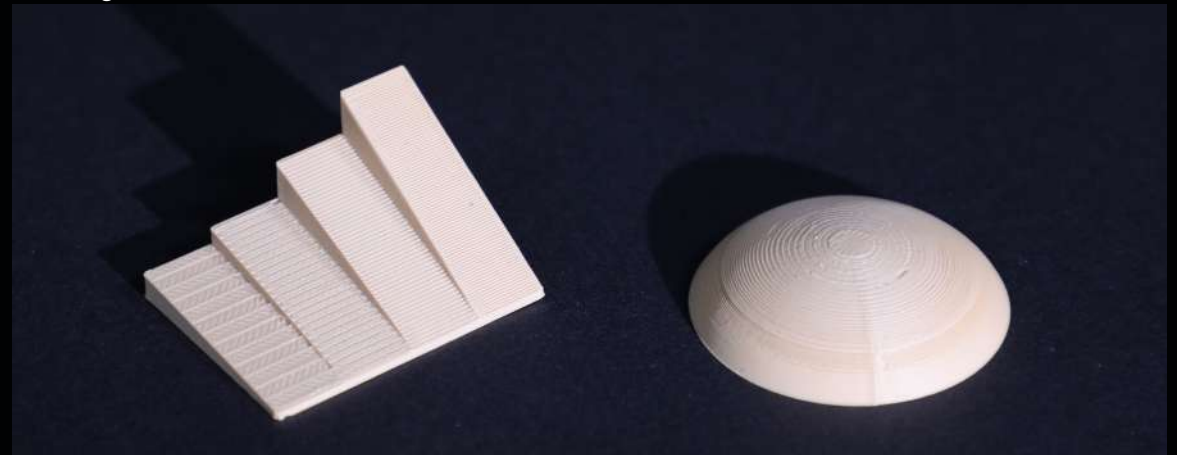


Fig 2.3 - Printed Object with ZAA On

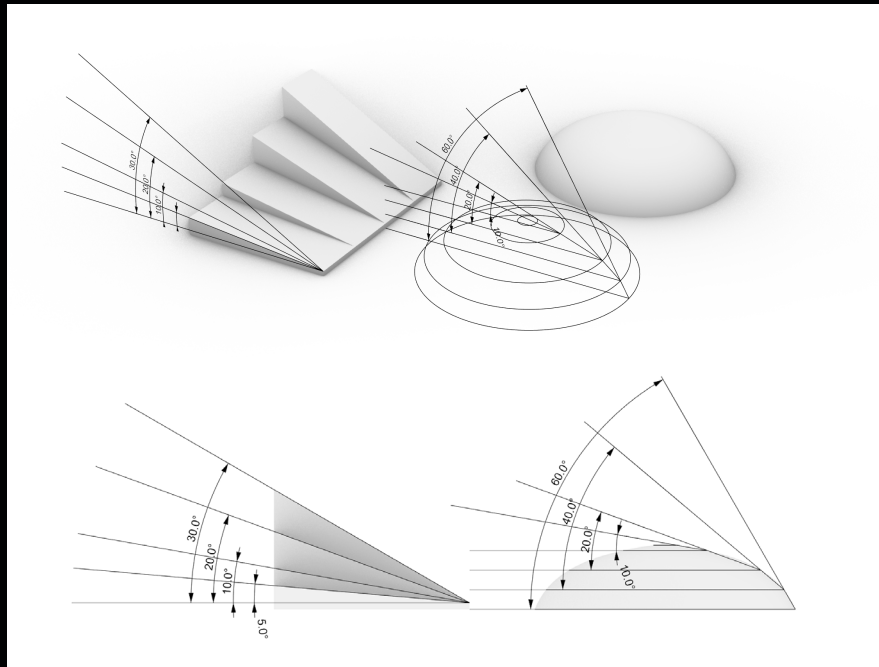


Fig 2.2 - Stair Test Object with 5°/10°/20°/30° Slope (Left); Dome Test Object with 10°/20°/40°/60° Steepness (Right)

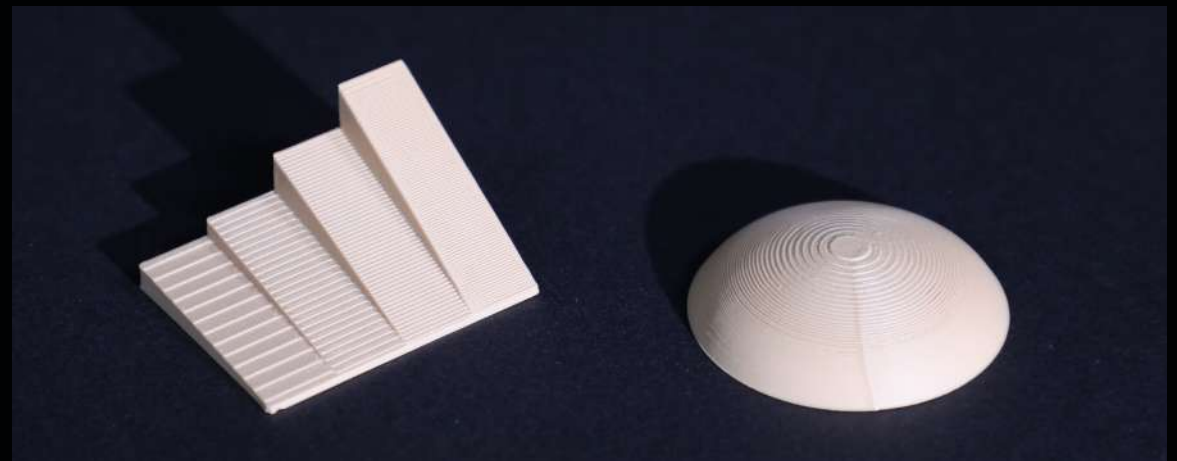


Fig 2.4 - Printed Object with ZAA Off

Bricklayer

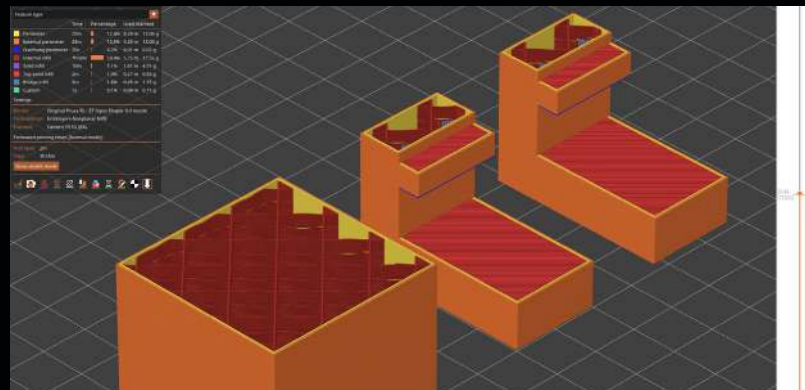


Fig 3.5.3.6 - Bricklayer Slicing - two different layers

Cantilever bending tests were performed by clamping the specimen, suspending a 3kg bucket, and incrementally adding 0.17kg metal threads until failure. Control failed at 4.02kg, GeekDetour at 3.68kg, and the non-planar variant at $\leq 3\text{kg}$ (bucket alone). Community discussions cite tensile improvements [12]; neither method improved bending strength under this protocol.



Fig 3.7 - Strength Test Setup - 3kg Metal Basket with 10mm thick handle

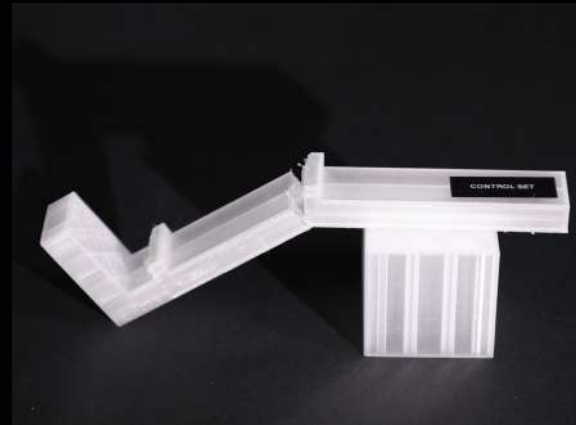


Fig 3.8 - Control Set Object after Strength Test

$$\begin{array}{c}
 \text{Bucket} \\
 3\text{kg}
 \end{array}
 +
 \begin{array}{c}
 \text{Threads} \\
 0.17 \times 6 = 1.02\text{kg}
 \end{array}
 = 4.02\text{kg}$$

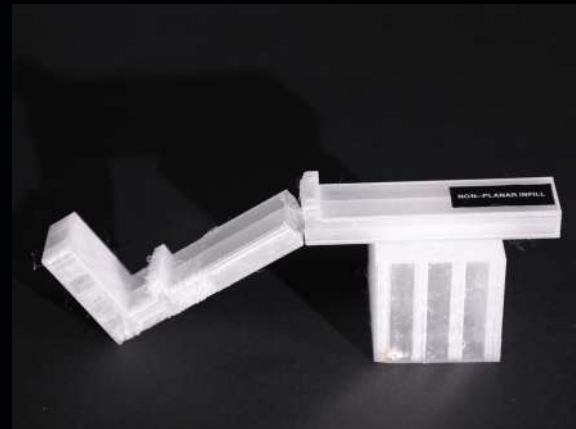


Fig 3.9 - Non-Planar Infill Object after Strength Test

$$\begin{array}{c}
 \text{Bucket} \\
 3\text{kg}
 \end{array}
 = 3\text{kg}$$

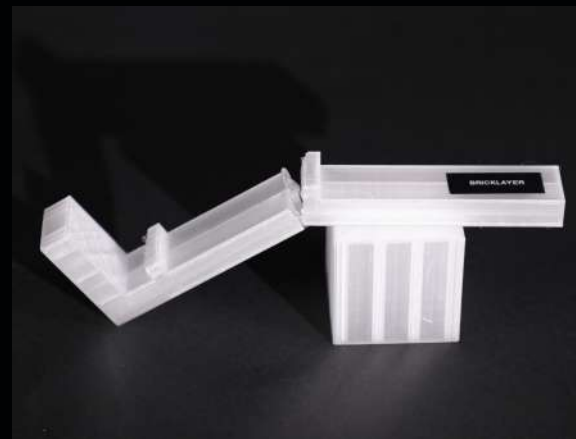


Fig 3.10 - Bricklayer Object after Strength Test

$$\begin{array}{c}
 \text{Bucket} \\
 3\text{kg}
 \end{array}
 +
 \begin{array}{c}
 \text{Threads} \\
 0.17 \times 4 = 0.68\text{kg}
 \end{array}
 = 3.68\text{kg}$$

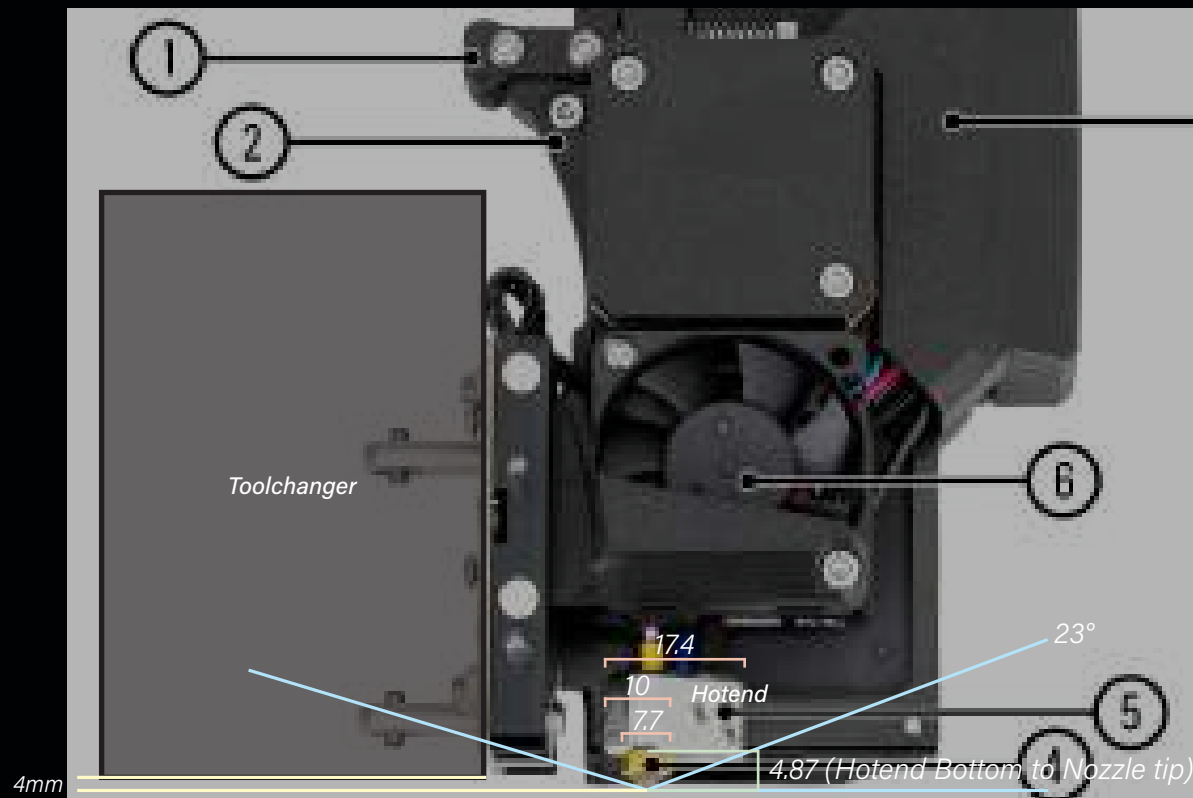
Strength Test Full Operation



04

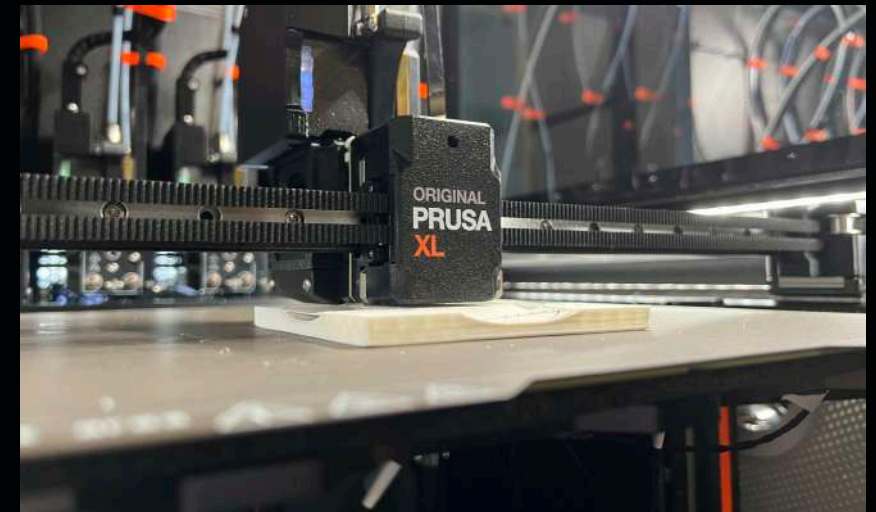
Prusa Non-Planar Slicing

True non-planar slicing was performed via the EiNSTein- PrusaSlicer 2.6 fork [1], derived from Ahlers' Slic3r implementation [5] and demonstrated by TeachingTech [6]. Nextruder geometry on the Prusa XL 5T was measured; trigonometric analysis Maximum non-planar angle = 23° and Maximum non-planar height = 4mm. Recommended slicer settings were applied: fan off, reduced temperature, conservative speed, avoid-crossing-perimeter enabled, top layers sufficient to mask underlying stair-stepping. Undulating-block.stl was printed at progressive parameters ($10^\circ/3.5\text{mm}$, $20^\circ/3.5\text{mm}$, $23^\circ/4\text{mm}$). Minor Nextruder contact at 4mm produced surface bumps; the working envelope was revised to $23^\circ/3.7\text{mm}$.



TeachingTech Recommended Setting for Non-Planar Slicing [6]

- No Cooling
- Reduce Nozzle Temperature by 20 Celsius
- Disable Idle Temperature
- 40-60mm/s Speed
- Enable Avoid crossing perimeters
- Keep 3-5 Top Layer for better result



Max. Non-Planar Height: 4mm (Bottom of Toolchanger to Nozzle tip)

Max. Non-Planar Angle: 23° (Hotend End to Nozzle tip) = $\tan^{-1}(4.87/11.25)$

Half Nozzle Width = $3.85 = 7.7/2$

Hotend Short Side to Nozzle tip = $6.15 = 10 - 3.85$

Hotend Long Side to Nozzle tip = $11.25 = 17.4 - 6.15$



Moment of printing
Undulating Block

Fig 4.1 - Slicing of Control Set (Non-Planar Off)

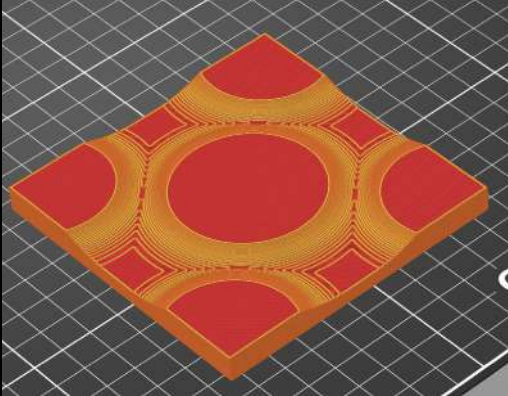


Fig 4.2 - Slicing of Non-Planar Layer (10°/3.5mm)

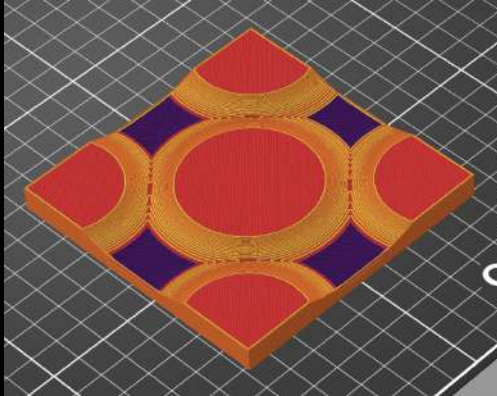


Fig 4.3 - Slicing of Non-Planar Layer (20°/3.5mm)

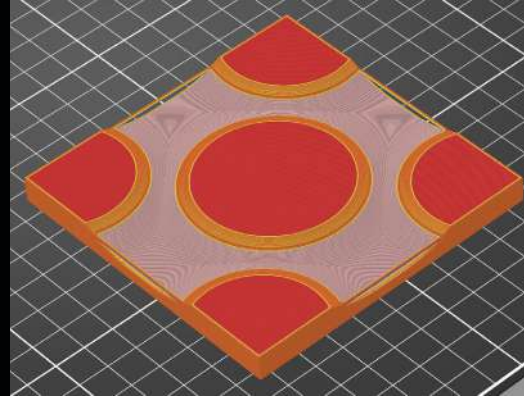
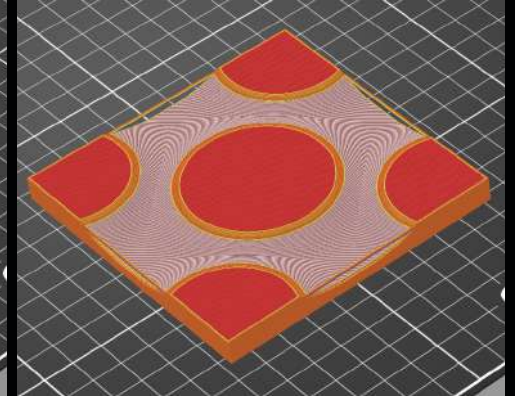


Fig 4.4 - Slicing of Non-Planar Layer (23°/4mm)



Non-Planar Layer Toolpath

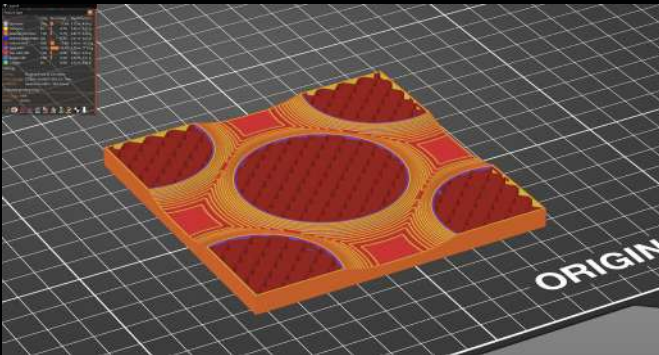


Fig 4.5 - Layers before Non-Planar Layers

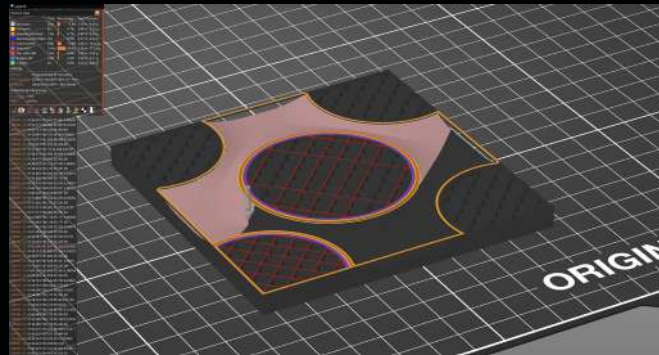


Fig 4.6 - First Non-Planar Layer Toolpath

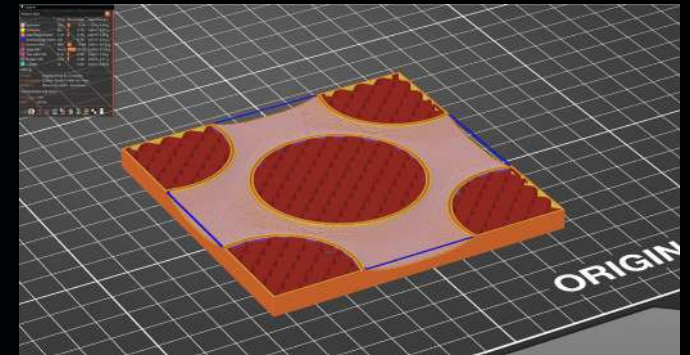


Fig 4.7 - First Non-Planar Layer

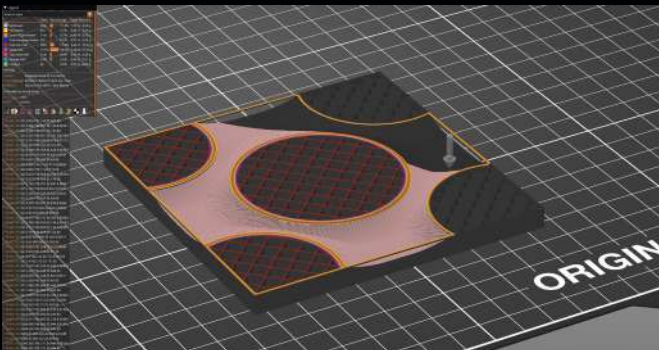


Fig 4.8 - Following Non-Planar Layer Toolpath

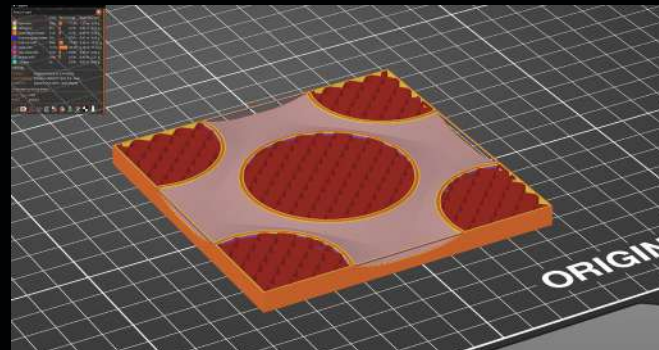


Fig 4.9 - Following Non-Planar Layer (Opposite Diagonal)

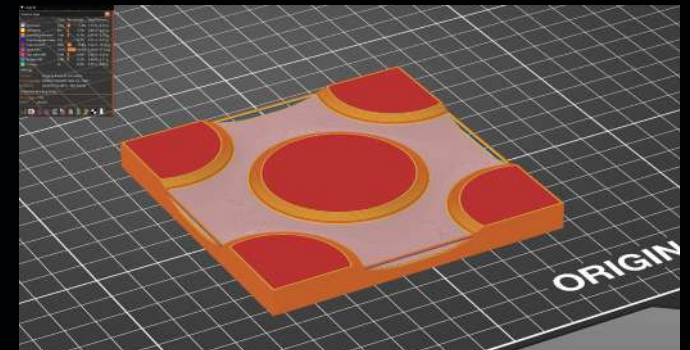


Fig 4.10 - Layers after Non-Planar Layers



Fig 4.11 - Control Set (Non-Planar Off); Printing Time 2h25m



Fig 4.12 - Non-Planar Slicing (10°/3.5mm); Printing Time 2h33m



Fig 4.13 - Non-Planar Slicing (20°/3.5mm); Printing Time 2h47m



Fig 4.14 - Non-Planar Slicing (23°/4mm); Printing Time 2h50m



Moment of printing Non-Planar Layer (10°/3.5mm)



Moment of printing Non-Planar Layer (20°/3.5mm)



Moment of printing Non-Planar Layer (23°/4mm)



Moment of Manually Remove Elevated Vertical Perimeter

05

Transparency Test - Non Planar Slicing

Non-planar slicing was evaluated as a transparency enhancement on the Prusa XL 5T in translucent PETG. Simple-block.stl was printed with CNC Kitchen's transparency settings [7], top/bottom layers retained for non-planar viability. Infill varied (15%, 50%) with and without non-planar processing. The non-planar variable produced negligible transparency difference within each pair. Infill density dominated optical clarity; surface treatment was secondary.

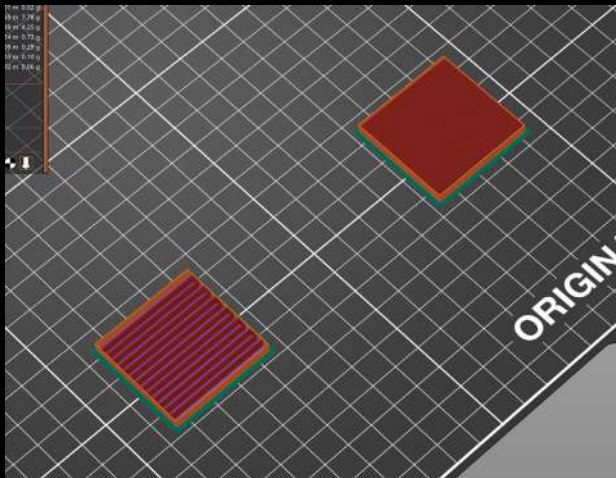


Fig 5.1 - 15% infill (Left) and 50% infill (Right)

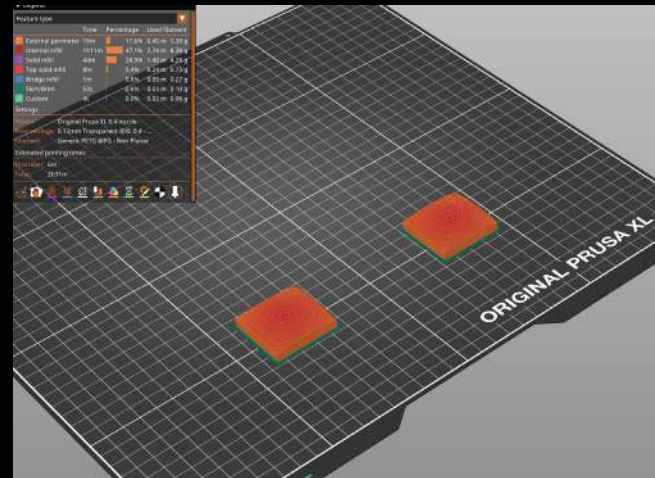


Fig 5.2 - Slicing of Control Set (2h31m)

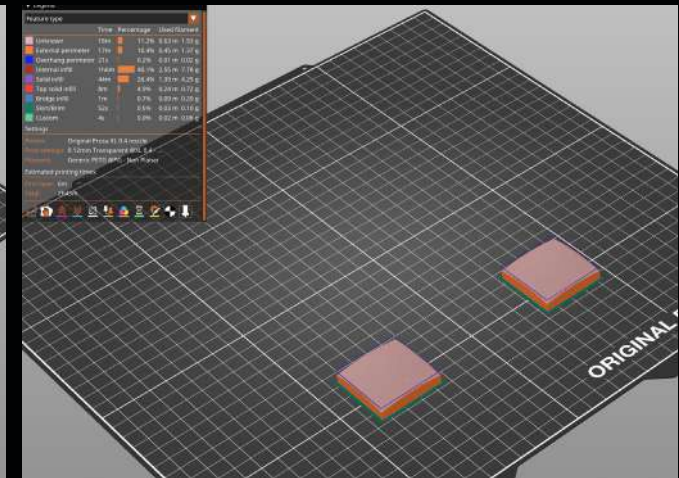


Fig 5.3 - Slicing with Non-planar enable (2h45m)

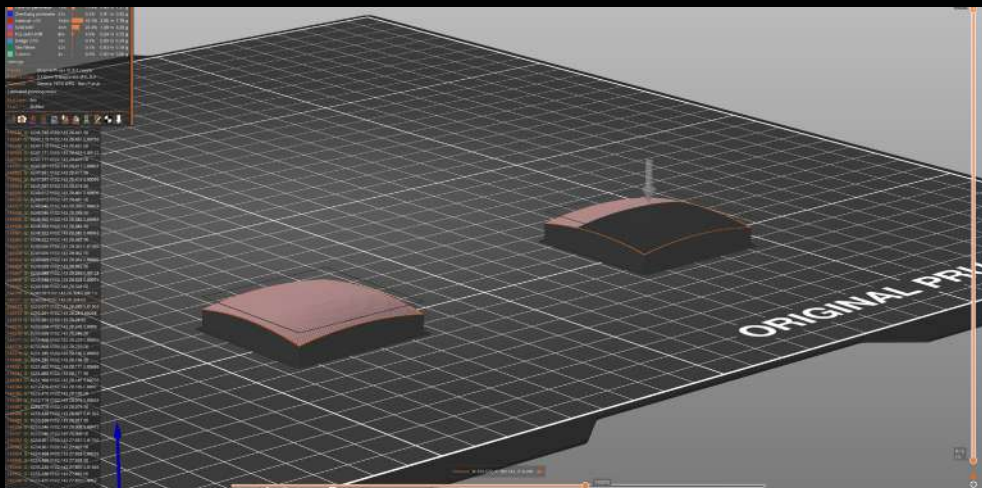


Fig 5.4 - Non-planar toolpath on Simple-block.stl

CNC Kitchen Recommended Setting for Best Transparent Result [7]

- 0.12mm Layer Height
- 1 Perimeter Vertical Wall
- Aligned Rectilinear infill with 0 fill angle
- 220 Celsius for translucent PETG
- No cooling
- 15mm/s Speed
- 102% Extrusion Rate
- 100% infill density (did not Apply)
- 0 Top + Bottom Layer (did not Apply)



Fig 5.5 - Transparency Test Print Left to Right: Non-Planar Print (15% infill); Non-Planar Print (50% infill); Control Set Print (15% infill); Control Set Print (50% infill)

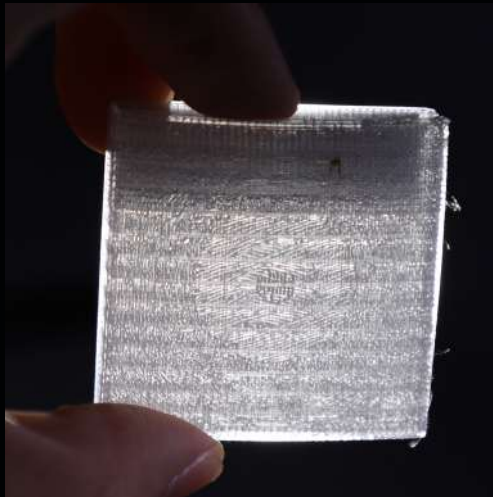


Fig 5.6 - Non-Planar Slicing Print (15% infill)

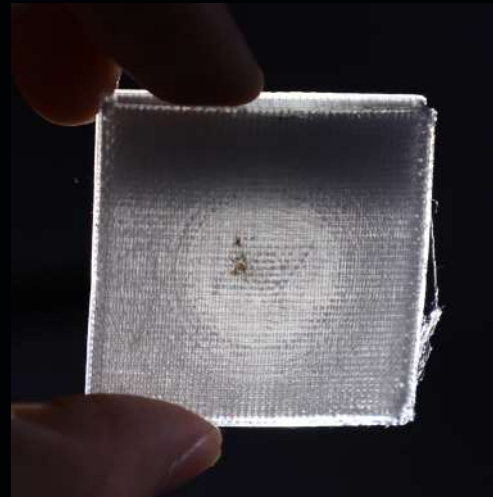


Fig 5.7 - Non-Planar Slicing Print (50% infill)

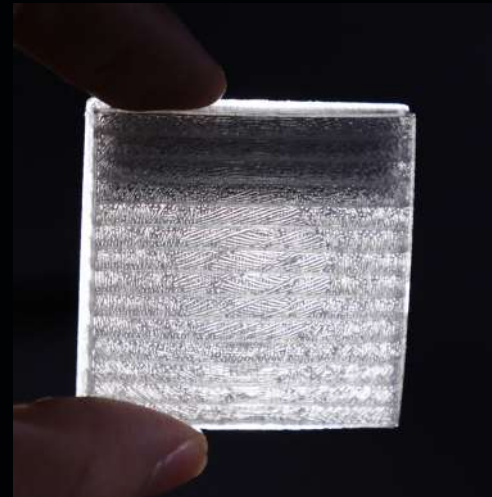


Fig 5.8 - Control Set Print (15% infill)

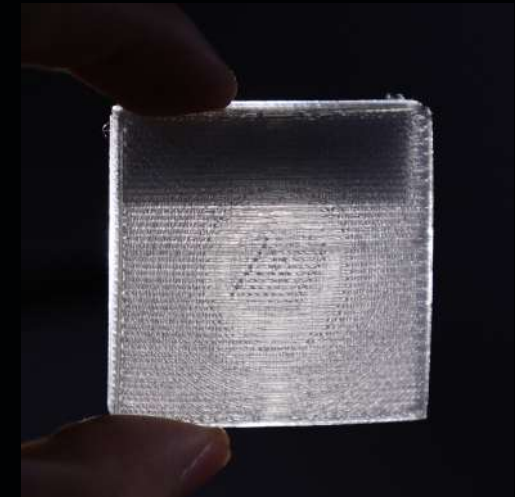


Fig 5.9 - Control Set Print (50% infill)



Moment of printing
the First PETG layer



Moment of printing
15%/50% infill



Moment of printing Stair
Steeping Layer
(Start Losing Transparency)



Moment of printing
Non-Planar Surface

06

Application - Insole

Cendrero et al. [8] demonstrated non-planar FDM insole production on a 3-axis machine, validating the insole as a non-planar case study. The geometry was tested within the DRH envelope: a base by punkti [9] was re-modelled in Rhino 3D to 3.7mm maximum variation and printed in TPU as control and non-planar (23°/3.7mm) sets.

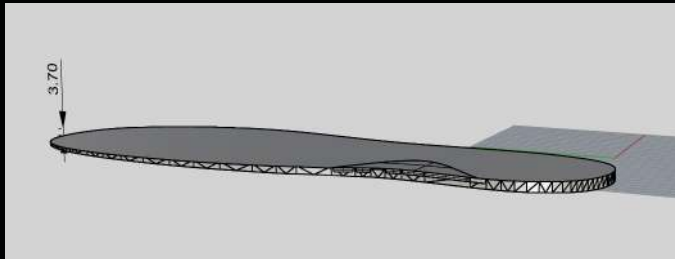


Fig 6.1 - Rhino Model of Insole with 3.7mm height of steepness

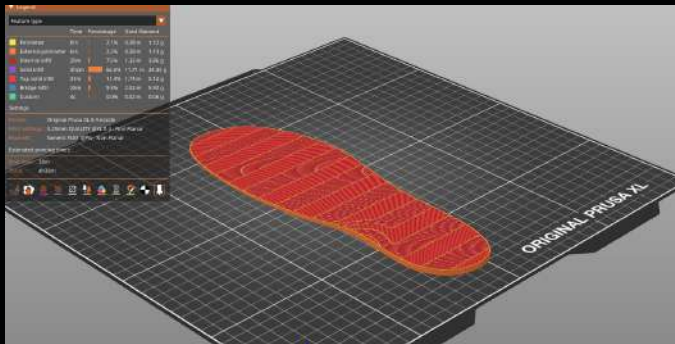


Fig 6.2 - Control Set Slicing (Insole) 4h33m

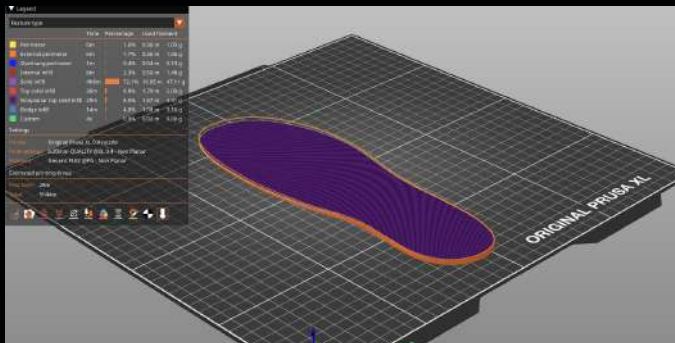


Fig 6.3 - Non-Planar Slicing (Insole) 5h44m



Fig 6.4 - Control Set Printed Object (Insole) 4h33m



Fig 6.5 - Non-Planar Printed Object (Insole) 5h44m



Moment of printing
the First TPU layer



Moment of printing
Non-Planar Surface



Moment of manually
Remove Elevated
Vertical Perimeter

Application - Grasshopper Parametric Non-Planar Surface Generator

A Grasshopper script generated parametric sine surfaces following Wang [10], constrained by 3.7mm non planar height and non-planar angle equation:

$$T = \frac{\tan(\theta) \cdot L}{\pi}$$

The Graph Mapper sine output (0/1) introduced the π factor; angle-adjustment was coupled to Angle, as the relation maps poorly to Graph Mapper geometry. The control showed visible stair-stepping; the non-planar variant produced a continuous reflective surface in metallic black filament.



Fig 6.6 - Non-Planar Printed Object (Grasshopper Generated Surface)



Fig 6.7 - Control Set Printed Object (Grasshopper Generated Surface)

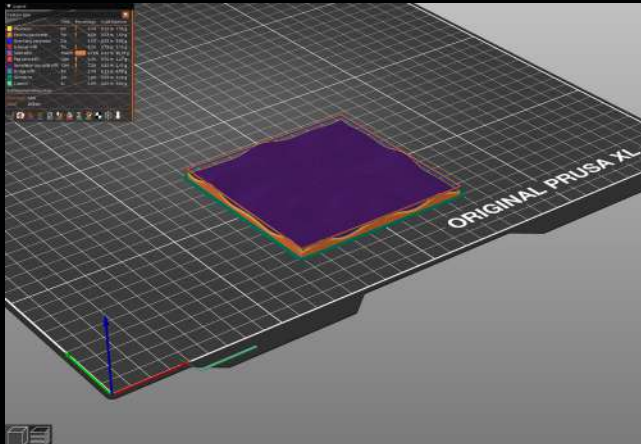


Fig 6.8 - Non-Planar Slicing (Grasshopper Generated Surface)

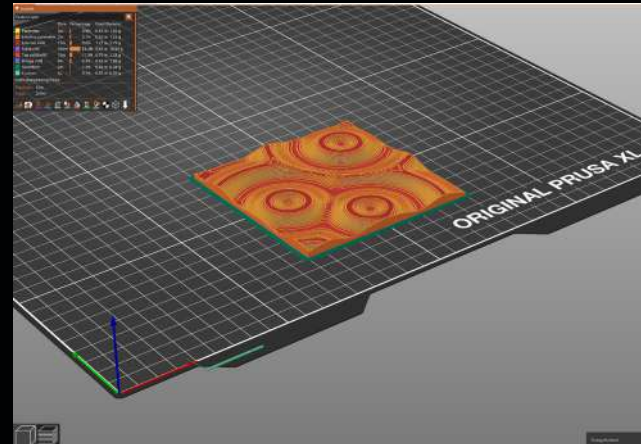


Fig 6.9 - Control Set Slicing (Grasshopper Generated Surface)

Moment of printing Grasshopper Generated
Non-Planar Surface



Moment of manually Remove Elevated
Vertical Perimeter during mid-print



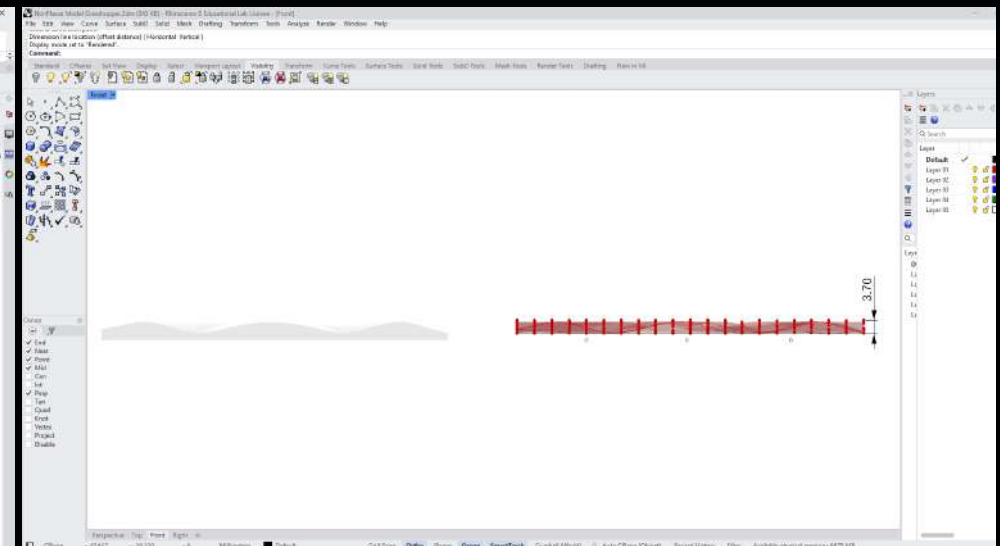


Fig 6.11 - Rhino View shows Grasshopper control the Maximum Non-Planar Height of Surface

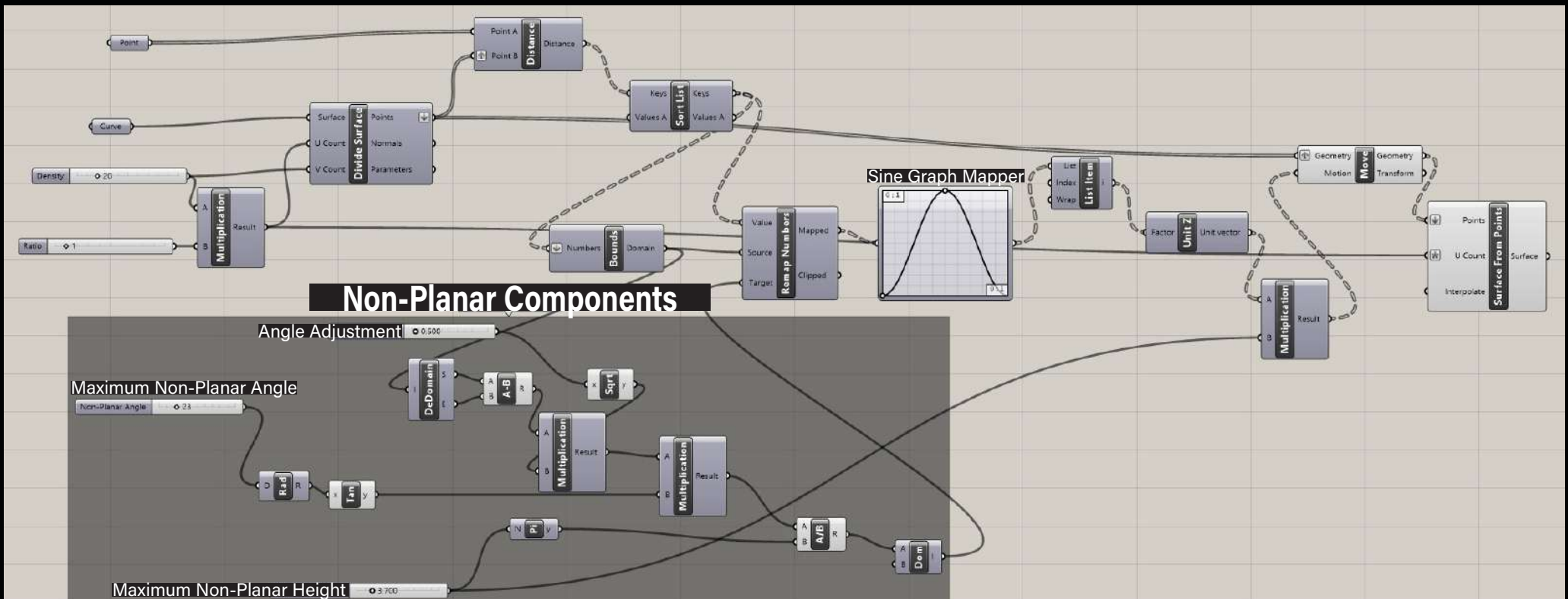


Fig 6.12 - Grasshopper View of Generated Parametric Non-Planar Surface with Customise components to adjust Non-Planar Parameter (inspired by Daoru Wang's Parametric Vase Grasshopper Tutorial)

07

Conclusion

Fig 7.1 - Elevated vertical wall on Insole mid-print



Fig 7.2 - Vertical perimeter string Overlay by Top Layer



Software-only non-planar slicing on the Prusa XL 5T proved viable within a 23°/3.7mm envelope, producing smooth surfaces on shallow ergonomic geometries but limited by Nextruder collision risk [11] and slicer instability. Infill density dominated transparency over surface treatment. Persistent issues included elevated vertical perimeter strings requiring manual removal by hand mid-print, version lock to PrusaSlicer 2.6, and orientation-dependent failures. Future work: porting to OrcaSlicer, hardware modification, and resolving the perimeter bug.

Hardware Modification Options



Fig 7.3 - Bending Nozzle from Non-planar Test[11]



Fig 7.4 - Airbrush Nozzle, risk of leaking filament



Fig 7.5 - Original Prusa ObXidian Nozzle - Nextruder Compatible (0.4mm, 0.6mm) for (MK4/XL)



Fig 7.6 - Nextruder V6 Nozzle Adapter



Fig 7.7 - E3D Revo Six Hot End (Prusa Edition)



Fig 7.8 - V6 Volcano Brass Length Extruder Nozzle

References

- [1] Chagnon, F. (EiNSTeiN-). PrusaSlicer non-planar fork. github.com/EiNSTeiN-/PrusaSlicer
- [2] adob. BambuStudio-ZAA. github.com/adob/BambuStudio-ZAA
- [3] GeekDetour. BrickLayers. github.com/GeekDetour/BrickLayers
- [4] TengerTechnologies. Bricklayers. github.com/TengerTechnologies/Bricklayers
- [5] Ahlers, D. 3D Printing of Nonplanar Layers for Smooth Surface Generation.
- [6] TeachingTech. Non-planar 3D printing tutorial. youtube.com/watch?v=cbhWni9f980
- [7] CNC Kitchen. Transparent FDM 3D printing. youtube.com/watch?v=9qb25Gi4Jv0
- [8] Cendrero, A.M.; Fortunato, G.M.; Munoz-Guijosa, J.M.; De Maria, C.; Díaz Lantada, A. Benefits of Non-Planar Printing Strategies Towards Eco-Efficient 3D Printing. Sustainability 2021, 13, 1599.
- [9] punkti. Scarpa F1 Insole Spacer. makerworld.com/en/models/945524
- [10] Wang, D. Grasshopper parametric surface tutorial. youtube.com/watch?v=MzDwjAEmavU
- [11] r/prusa3d community discussion. Prusa XL non-planar printing risks. reddit.com/r/prusa3d/comments/1qq8kj8